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NIMCET-Free Test



By Amit Katiyar (MCA-JNU)



1. A student calculated mean and standard deviation of 100 observation as 40 and 51 respectively. Later it was discovered that he was taken by mistake 50 instead of 40, then new standard deviation is:

NIMCET 2007

- (a)5 (b)4.9
(c)5.1 (d)none

2. The average marks per student in a class of 30 students is 45. On rechecking it was found that marks had been entered wrongly in two cases. After correction these marks were increased by 24 and 34 in the two cases. The correct average marks per student are

NIMCET 2010

- (a)75 (b)60 (c)56 (d)47

3. Two variables have least square regression lines $3x + 2y = 26$ and $6x + y = 31$, then correlation between x and y is

NIMCET 2011

- (a)0.5 (b)0.7 (c)-0.7 (d)-0.5

4. Suppose values taken by a random variable X are such that $a \leq x_1 \leq b$, where x_i denotes the value of X in the i th case for $i = 1, 2, 3, \dots, n$ then

NIMCET 2012

- (a) $(b - a)^2 \geq \text{Var}(X)$ (b) $\frac{a^2}{4} \leq \text{Var}(X)$
(c) $a^2 \leq \text{Var}(X) \leq b^2$ (d) $a \leq \text{Var}(X) \leq b$

5. If $\bar{x}_1$ and $\bar{x}_2$ are the means of two distributions such that $\bar{x}_1 < \bar{x}_2$ and $\bar{x}$ is the mean of the combined distribution, then

NIMCET 2013

- (a) $\bar{x} < \bar{x}_2$ (b) $\bar{x} > \bar{x}_2$
(c) $\bar{x} = \frac{\bar{x}_1 + \bar{x}_2}{2}$ (d) $\bar{x}_1 < \bar{x} < \bar{x}_2$

6. A random variable x has the distribution law as given below

NIMCET 2013

x	1	2	3
$P(X) = x$	0.3	0.4	0.3

The variance of the distribution is

- (a)0.4 (b)0.6 (c)2 (d)None

7. In a group of 200 students, the mean and standard deviation of scores were found to be 40 and 15, respectively. Later on it was found that two scores 43 and 35 were misread as 34 and 53, respectively. The corrected means of scores is

NIMCET 2014

- (a)40.95 (b)39.0 (c)39.95 (d)43

8. The mean deviation from the mean of the AP $a, a + d, a + 2d, \dots, a + 2nd$ is

NIMCET 2014

- (a) $\frac{n}{n+1}d$ (b) $\frac{n(n+1)}{2n(n+1)}d$ (c) $\frac{n+1}{2n+1}d$ (d) $\frac{n(n+1)}{2n+1}d$

9. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is

NIMCET 2014

- (a)80% (b)60% (c)40% (d)20%

10. If the mean deviation of the number $1, 1 + d, 1 + 2d, \dots, 1 + 100d$ from their mean is 225, then the value of d is

NIMCET 2015

- (a)20.0 (b)10.1 (c)20.2 (d)10.0

11. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is

NIMCET 2016

- (a)80% (b)60% (c)40% (d)20%

12. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observation 151, 152, ..., 250. If V_A and V_B represent variance of the two population respectively, then $\frac{V_A}{V_B}$ is

NIMCET 2016

- (a) $\frac{9}{4}$ (b) $\frac{4}{9}$ (c)1 (d) $\frac{2}{3}$

13. Let $x_1, x_2, \dots, x_n$ be n observation such that $\sum x_i^2 = 400$ and $\sum x_i = 80$. Then a possible value of n among the following is

NIMCET 2016

- (a)10 (b)15 (c)20 (d)8

14. If $\bar{x}$ is the mean of distribution of X , then usual notation $\sum_{i=1}^n f_i(x_i - \bar{x})$ is

NIMCET 2017

- (a) Mean deviation about mean
(b) Standard deviation
(c) 1 (d) 0

15. Let $x_i, i = 1, 2, \dots, n$ be n observations and $w_i = px_i + k, i = 1, 2, \dots, n$ where p and k are constants. If the mean of x_i 's is 48 and standard deviation is 12, whereas the mean of w_i 's is 55 and standard deviation is 15, the values of p and k should be

NIMCET 2019

- (a) $p = 1.25, k = -5$ (b) $p = -1.25, k = 5$
(c) $p = 2.5, k = -5$ (d) $p = 25, k = 5$

16. If the mean deviation of the numbers $1, 1 + d, 1 + 2d, \dots, 1 + 100d$ from their mean is 255, then d is equal to

NIMCET 2019

- (a)10.1 (b)10.2 (c)10.3 (d)10.4

17. The mean of 5 observations is 5 and their variance is 12.4. if three of the observations are 1, 2, 6, then the mean deviation from the mean of the data is

NIMCET 2019

- (a)2.5 (b)2.6 (c)2.8 (d)2.4

18. A, B, C are three sets of values of x
A: 2,3,7,1,3,2,3, B:7,5,9,1,2,5,3,8, C:4,4,11,7,2,3,4
Select the correct statement among the following

NIMCET 2020

- (a) Mean of A is equal to Mode of C
(b) Mean of C is equal to Median of B
(c) Median of B is equal to Mode of A
(d) Mean, Median and Mode of A are same



19. Standard deviation for the following distribution is

Size of item	6	7	8	9	10	11	12
Frequency	3	6	9	13	8	5	4

NIMCET 2020

- (a) 1.607 (b) 9.0 (c) 5.0 (d) 1.88

20. If a variable takes value 0, 1, 2, 3,, 50 with frequency $1, {}^{50}C_1, {}^{50}C_2, \dots, {}^{50}C_{50}$ then the A.M. is

NIMCET 2021

- (a) 50 (b) 25 (c) ${}^{50}C_2$ (d) 51

21. The standard deviation of 20 numbers is 30. If each of the numbers is increased by 4, then the new standard deviation will be

NIMCET 2021

- (a) 24 (b) 34 (c) 30 (d) 20

22. Consider the following frequency distribution table.

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	180	f_1	34	180	136	f_2	50

If the total frequency is 686 and median is 42.6 then the values of f_1 and f_2 are

NIMCET 2021

- (a) 81, 25 (b) 82, 24 (c) 83, 23 (d) 84, 22

23. The first three moments of a distribution about 2 are 1, 16, 40 respectively. Then mean and variance of the distribution are

NIMCET 2022

- (a) (2, 16) (b) (2, 15) (c) (3, 15) (d) (1, 16)

24. The 10th and 50th percentiles of the observations 32, 49, 23, 29, 118 respectively are

NIMCET 2022

- (a) 21, 32 (b) 23, 32
(c) 23, 33 (d) 22, 31

25. The mean of 25 observations was found to be 38. It was later discovered that 23 and 38 were misread as 25 and 36, then the mean is

NIMCET 2022

- (a) 32 (b) 36 (c) 38 (d) 42

26. The mean of 5 observation is 5 and their variance is 124. If three of the observations are 1, 2 and 6, then the mean deviation from the mean of the data is

NIMCET 2023

- (a) 2.5 (b) 2.6 (c) 2.8 (d) 2.4

27. For a group of 100 candidates, the mean and standard deviation of scores were found to be 40 and 15 respectively. Later on, it was found that the scores 25 and 35 were misread as 52 and 53 respectively. Then the corrected figures are

NIMCET 2023

- (a) 39.9, 14.97 (b) 39.5, 14
(c) 39.55, 14.97 (d) 40.19, 15.1

28. Consider the following frequency distribution table.

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	180	f_1	34	180	136	f_2	50

If the total frequency is 686 and median is 42.6 then the values of f_1 and f_2 are

NIMCET 2023

- (a) 80, 25 (b) 83, 22 (c) 79, 26 (d) 82, 23

29. It is given that the mean, median and mode of a data set is 1, 3^x and 9^x respectively. The possible values of the mode is (Statistics)

NIMCET 2024

- (a) 1, 4 (b) 1, 9
(c) 3, 9 (d) 9, 8

30. Given a set A with median $m_1 = 2$ and set B with median $m_2 = 4$ What can we say about the median of the combined set? (Statistics)

- (a) at most 1 (b) at most 2
(c) at least 1 (d) at least 2

NIMCET 2024

31. There are 40 female and 20 male students in a class. If the average heights of female and male students are 5.15 feet and 5.66 feet, respectively, then the average height (in feet) of all the students in the class equals

NIMCET 2025

- (a) 5.490 (b) 5.405
(c) 5.175 (d) 5.320

32. Which one of the following is **NOT** a correct statement?

NIMCET 2025

- (a) The standard deviation is greater than or equal to the mean deviation (about mean)
(b) The variance is expressed in the same units as the units of observation
(c) The value of standard deviation changes by a change of scale.
(d) The sum of squares of deviations is minimum when taken from the mean

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	d	d	a	d	b	c	d	a	b
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
a	c	c	d	a	a	c	d	a	b
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
c	b	c	b	c	c	c	d	a	d
31.	32.								
d	b								

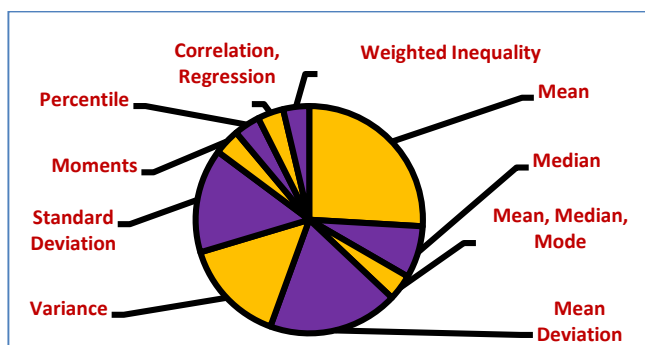
Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	01	NIMCET 2016	03
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	00	NIMCET 2018	00
NIMCET 2010	01	NIMCET 2019	03
NIMCET 2011	01	NIMCET 2020	02
NIMCET 2012	01	NIMCET 2021	03
NIMCET 2013	02	NIMCET 2022	03
NIMCET 2014	03	NIMCET 2023	03
NIMCET 2015	01	NIMCET 2024	02
		NIMCET 2025	02



SUB TOPICWISE ANALYSIS



Sub Topic	Qs. Asked
Mean	07
Median	02
Mean, Median, Mode	01
Mean Deviation	05
Variance	04
Standard Deviation	04
Moments	01
Percentile	01
Correlation, Regression	01
Weighted Inequality	01

SOLUTION

1. (d)

$$n = 100$$

$$\bar{x} = 40$$

$$\sigma = 51$$

$$\text{Incorrect value} = 50$$

$$\text{correct value} = 40$$

$$\text{New mean} = 40 - \frac{50-40}{100}$$

$$r^2 = \frac{\sum x_i^2}{100} - (40)^2 \left\{ \because \theta^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2 \right\}$$

$$(51)^2 + (40)^2 = \frac{\sum x_i^2}{100}$$

$$\sum x_i^2 = 420,100$$

$$\text{New } \sum x_i^2 = 420,100 + (40)^2 - (50)^2 = 419,200$$

$$\text{New SD} = \sqrt{\frac{\sum (x_i)^2}{100} - (399)^2}$$

$$= \sqrt{\frac{419200}{100} - 1592.01}$$

$$\text{SD} = 50.99$$

$$\sigma = 51$$

2. (d)

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_{30}}{30} = \frac{\sum_{i=1}^{30} x_i}{30}$$

$$\bar{x} = 45 \text{ (given)}$$

$$45 \times 30 = \sum_{i=1}^{30} x_i = 1350$$

$$\text{Actual sum of marks of 30 student}$$

$$= 1350 + 24 + 34 = 1408$$

$$\text{Average} = \frac{1408}{30} = 46.93 = 47$$

3. (d) Let regression line y on x

$$3x + 2y = 26, 2y = -3x + 26$$

$$= -\frac{3}{2}x + 13 \quad b_{yx} = -3/2$$

Let regression line x on y

$$6x + y = 31$$

$$x = \frac{-y}{6} + \frac{31}{6} \quad b_{xy} = \frac{-1}{6}$$

$$r = \sqrt{b_{yx}b_{xy}} = \sqrt{-\frac{3}{2} \times \frac{-1}{6}} = \sqrt{0.25} = -0.5$$

$\therefore$ sign of r , b_{xy} and b_{yx} is always same.

so $r = -0.5$.

4. (a) $a \leq x_i \leq b$

$$\text{Range} = \text{Maximum} - \text{Minimum} = b - a$$

We should know Standard deviation $\leq$ Range

$$\sigma \leq (b - a)$$

$$\text{Square both side } \sigma^2 \leq (b - a)^2$$

We should know the variance = (Standard deviation)²

5. (d) Given $\bar{x}_1$ and $\bar{x}_2$ are two means of distributions and also $\bar{x}_1 \leq \bar{x}_2$ and $\bar{x}$ is the combined mean of the distribution. Here, we assume that both distributions are of equal size.

$$\bar{x} = \frac{\bar{x}_1 + \bar{x}_2}{2}$$

$$\bar{x}_1 < \bar{x}_2$$

$$2\bar{x}_1 < \bar{x}_1 + \bar{x}_2$$

$$\text{Hence, } \frac{\bar{x}_1 + \bar{x}_2}{2} > \bar{x}_1 \Rightarrow \bar{x} > \bar{x}_1$$

$$\text{Similarly, } \bar{x}_2 > \bar{x}_1$$

$$2\bar{x}_2 > \bar{x}_1 + \bar{x}_2$$

$$\text{Hence, } \frac{\bar{x}_1 + \bar{x}_2}{2} < \bar{x}_2 \Rightarrow \bar{x} < \bar{x}_2$$

$$\text{So, } \bar{x}_1 < \bar{x} < \bar{x}_2$$

6. (b) Variance = $\sum (x_i - \bar{x})^2 P_i$

$$\text{Here, } \bar{x} = \sum x_i P_i = 1 \times 0.3 + 2 \times 0.4 + 3 \times 0.3 = 2$$

Variance

$$= (1-2)^2 \times 0.3 + (2-2)^2 \times 0.4 + (3-2)^2 \times 0.3 = 0.6$$

7. (c)

$$\bar{x} = \frac{\sum x_i}{n} \Rightarrow \sum x_i = n\bar{x} = 200 \times 40 = 8000$$

$$\text{Correct } \sum x_i = \frac{8000 - 34 - 53 + 43 + 35}{200}$$

$$= \frac{7991}{200} = 39.955 \text{ (Correct mean)}$$

8. (d) $a, a + d, a + 2d, \dots, a + 2nd$

$$a + (N-1)d = 2 + 2nd$$

$$N-1 = 2n$$

$$\text{Total number of terms } N = 2n + 1$$

Hence, $a + 2nd$ is $(2n+1)$ th term of AP.

$$\text{Mean} = \frac{(a) + (a+d) + (a+2d) + \dots + (a+2nd)}{2n+1}$$

$$= \frac{\left(\frac{2n+1}{2}\right)(a+a+2nd)}{2n+1}$$

$$\text{Mean} = a + nd$$

Mean deviation from mean

$$= \frac{1}{2n+1} \sum_{i=0}^{2n} |x_i - \bar{x}|$$



$$= \frac{1}{2n+1} [nd + (n-1)d + (n-2)d + \dots + d + 0]$$

$$= \frac{d}{2n+1} [n + (n-1) + (n-2) + \dots + 1 + 0 + 1 + \dots + (n-1) + n]$$

$$= \frac{d}{2n+1} [2(1 + 2 + 3 + \dots + n)] = \frac{2n(n+1)}{2} \cdot \frac{d}{(2n+1)}$$

$$= \frac{n(n+1)}{(2n+1)} \times d$$

9. (a) Number of boys = n_1 and number of girls = n_2

$$\bar{x}_1 = 52, \bar{x}_2 = 42 \Rightarrow \bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

$$50 = \frac{n_1 \times 52 + n_2 \times 42}{n_1 + n_2}$$

$$50n_1 + 50n_2 = 52n_1 + 42n_2 \Rightarrow 8n_2 = 2n_1 \Rightarrow n_1 = 4n_2$$

$$\% \text{ of boys} = \frac{n_1}{n_1 + n_2} \times 100 = 80\%$$

10. (b)

$$MD = \frac{d}{n} \cdot \frac{n^2 - 1}{4}$$

[n = no. of terms, d = common difference]

Apply here

AP: 1, 1+d, 1+2d, ..., 1+100d

So, n = 101

$$MD = \frac{d}{101} \cdot \frac{101^2 - 1}{4} = \frac{d}{101} \cdot \frac{10200}{4} = \frac{2550}{101} d$$

Given MD = 225:

$$\frac{2550}{101} d = 225 \Rightarrow d = \frac{225 \cdot 101}{2550}$$

$$d \approx 10$$

Final Ans. 10.

11. (a) Refer to solution 8.

12. (c) $\sigma_x^2 = \frac{\sum d_i^2}{n}$

Here, the deviations are taken from the mean.

Since A and B have both 100 consecutive integers, therefore both have same standard deviations and variance.

$$\therefore \frac{V_A}{V_B} = 1$$

(As $\sum d_i^2$ is same in both the case)

13. (c) $x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2 = 400$

$$x_1 + x_2 + x_3 + \dots + x_n = 80$$

AM of m^{th} power of terms > m^{th} power of AM

$$= \frac{x_1^m + x_2^m + x_3^m + \dots + x_n^m}{n} > \left(\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \right)^m$$

$$m = 2$$

$$= \frac{x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2}{n} > \left(\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \right)^2$$

$$= \frac{400}{n} > \frac{(80)^2}{n^2}$$

$$\Rightarrow n > \frac{6400}{400} \Rightarrow n > 16$$

So, 20 is answer.

14. (d) $\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} \Rightarrow \bar{x} \sum_{i=1}^n f_i = \sum_{i=1}^n f_i x_i$$

$$\sum_{i=1}^n f_i (x_i - \bar{x}) = \sum_{i=1}^n f_i x_i - \sum_{i=1}^n f_i \bar{x} = 0$$

15. (a) $w_i = px_i + k$ [$\bar{x} = 48$ and $\bar{w}_i = 55$]

$$\bar{w}_i = p\bar{x}_i + k$$

$$55 = p \times 48 + k \quad 48p + k = 55$$

$$S.D. (w_i) = p(S.D. x_i)$$

S.D. (w_i) is not affected by change of origin

$$15 = 12p \quad 48p + k = 55$$

$$p = 5/4 \quad 48 \times \frac{5}{4} + k = 55$$

$$p = 1.25 \quad k = 55 - 60 = -5$$

$$k = -5$$

16. (a) 1, 1 + d, 1 + 3d, 1 + 100d are in AP

a = 1, Total number of terms = 101

$$= \frac{n}{2} (a + l) = \frac{101}{2} (1 + 1 + 100d)$$

$$= \frac{101}{2} (2 + 100d) = 101(1 + 50d)$$

$$\text{Hence, mean of these numbers} = \frac{101(1+50d)}{101} = 1 + 50d$$

$$\text{Mean deviation from mean} = \frac{1}{101}$$

$$|1 - (1 + 50d)| + |1 + d - (1 + 50d)| + \dots + |100d - (1 + 50d)|$$

$$= \frac{2|d|}{101} \times \frac{50 \times 51}{2} = \frac{2550}{101} |d| = \frac{2550}{101} |d| = 255 \Rightarrow |d| = 10.1$$

17. (c) Let the two numbers be x and y.

Given, variance $\sigma^2 = 12.4$

Mean, $\bar{x} = 5$ and $n = 5$

$$\frac{1+2+6+x+y}{5} = 5$$

$$x + y = 16$$

Now,

$$12.4 = \frac{(1-5)^2 + (2-5)^2 + (6-5)^2 + (x-5)^2 + (y-5)^2}{5}$$

$$12.4 = \frac{16+9+1+(x-5)^2+(y-5)^2}{5}$$

$$(x-5)^2 + (16-x-5)^2 = 12.4 \times 5 - 26$$

$$(x-5)^2 + (11-x)^2 = 62 - 26$$

$$x^2 - 16x + 55 = 0$$

$$x^2 - 5x - 11x + 55 = 0$$

$$x(x-5) - 11(x-5) = 0$$

$$(x-5)(x-11) = 0$$

$$x = 11, 5$$

Hence, y = 11 or 5

So, mean deviation

$$= \frac{|1-5| + |2-5| + |6-5| + |5-5| + |11-5|}{5}$$

$$= \frac{4+3+1+0+6}{5} = \frac{14}{5} = 2.8$$

18. (d) For the data: A : 2, 3, 7, 1, 3, 2, 3

$$n = 7$$

$$\text{Mean} = \frac{2+3+7+1+3+2+3}{7} = \frac{21}{7} = 3$$

Arranging the data A in ascending order, we have

A: 1, 2, 2, 3, 3, 3, 7.

$$\text{Median} = \left(\frac{7+1}{2} \right)^{\text{th}} \text{ term} = 4^{\text{th}} \text{ term} = 3$$

Since, 3 access max. numbers of times, Mode = 3

Mean = Median = Mode

19. (a) Mean = $\frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$



$$\text{Variance} = \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sigma$$

$$\sum f = 3 + 6 + 9 + 13 + 8 + 5 + 4 = 48$$

$$\text{Then mean } \bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$\bar{x} = \frac{6 \times 3 + 7 \times 6 + 8 \times 9 + 9 \times 13 + 10 \times 8 + 11 \times 5 + 12 \times 4}{48}$$

$$\bar{x} = 9$$

$$\sigma^2 = \frac{\sum f_i x_i^2}{n} - \bar{x}^2$$

$$\therefore \frac{\sum f_i x_i^2}{n}$$

$$= 3 \times 6^2 + 6 \times 7^2 + 9 \times 8^2 + 13 \times 9^2 + 8 \times 10^2 + 5 \times 11^2 + 4 \times 12^2$$

$$\therefore \sigma^2 = 83.58 - (9)^2 = 2.58$$

$$\text{Standard deviation} = \sqrt{2.58} = 1.607$$

20. (b)

$$\bar{x} = \frac{0 \times 1 + 1 \cdot {}^{50}C_1 + 2 \cdot {}^{50}C_2 + \dots + 5 \cdot {}^{50}C_5}{1 + {}^{50}C_1 + {}^{50}C_2 + \dots + {}^{50}C_5}$$

$$= \frac{\sum_{r=0}^{50} r \cdot {}^{50}C_r}{\sum_{r=0}^{50} {}^{50}C_r} = \frac{50 \cdot 2^{49}}{2^{50}} = \frac{50}{2} = 25$$

21. (c)

Let the number $x_1, x_2, \dots, x_{20}$

$$\text{Mean}(\bar{x}) = \frac{x_1 + x_2 + \dots + x_{20}}{20}, \sigma(x) = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = 30$$

Let new numbers $y_1, y_2, y_3, \dots, y_{20}$

$$y_1 = x_1 + 4, y_2 = x_2 + 4,$$

$$y_3 = x_3 + 4, \dots, y_{20} = x_{20} + 4$$

$$\bar{y} = \frac{y_1 + y_2 + \dots + y_{20}}{20} = \bar{x} + 4$$

$$y_i - \bar{y} = (x_i + 4) - (\bar{x} + 4), \text{ where } i = 1, 2, 3, \dots, 20$$

$$y_i - \bar{y} = x_i - \bar{x}$$

$$\sigma(y) = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} = 30$$

Now, we know that if each term of data is either increased or decreased by the same constant number then the standard deviation of the data is remains unchanged. And in question its already given that the standard deviation of 20 numbers is 30.

22. (b)

Class	Frequency	Cumulative Frequency
10 - 20	180	180
20 - 30	f_1	$180 + f_1$
30 - 40	34	$214 + f_1$
40 - 50	180	$394 + f_1$
50 - 60	136	$530 + f_1$
60 - 70	f_2	$530 + f_1 + f_2$
70 - 80	50	$580 + f_1 + f_2$

$$\text{Total frequency} = 686$$

$$580 + f_1 + f_2 = 686$$

$$\Rightarrow f_1 + f_2 = 106$$

$$\text{Median class} = \frac{N}{2} = \frac{686}{2} = 343$$

Median = 42.6 (given) lies in (40 - 50) interval Class 40 - 50

$$\text{Median} = l + \frac{\frac{N}{2} - C \cdot f}{f} \times h \text{ when } l = 40, h = 10$$

$$C \cdot F = 214 + f_1$$

$$42.6 = 40 + \frac{\frac{686}{2} - (214 + f_1)}{180} \times 10$$

$$2.6 \times 18 = 343 - 214 - f_1$$

$$46.8 = 129 - f_1$$

$$\Rightarrow f_1 = 82.2 \approx 82$$

$$\text{Now } f_1 + f_2 = 106$$

Put the value of f_1

$$\Rightarrow f_2 = 106 - f_1$$

$$\Rightarrow f_2 = 106 - 82 = 24$$

23. (c)

$$\mu_r = \frac{\sum (x - \bar{x})^r}{n}, \mu_r = r\text{th}$$

Moment about mean and $\sum (x - \bar{x}) = \text{Mean deviation}$

$$\mu_1 = \frac{\sum (x - \bar{x})}{n}; \mu_2 = \frac{\sum (x - \bar{x})^2}{n}; \mu_3 = \frac{\sum (x - \bar{x})^3}{n}$$

$$\mu_1 \text{ about 2 is 1, } \mu_2 \text{ is 16; } \mu_3 = 40$$

$$\mu_1 = \frac{\sum (x_i - 2)}{n} = 1 \quad \mu_2 = \frac{\sum (x_i - 2)^2}{n} = 16$$

$$\sum_{i=1}^n (x_i - 2) = n \quad \sum (x_i^2 + 4 - 4x_i) = 16n$$

$$\sum_{i=1}^n x_i - 2 \sum_{i=1}^n 1 = n \quad \sum x_i^2 - 4 \sum x_i + 4n = 16n$$

$$\sum x_i = 3n \Rightarrow \text{mean } \frac{\sum x_i}{n} = 3 \quad \sum x_i^2 = 24n$$

$$\text{Variance} = (\text{SD})^2 \cdot V(x) = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2$$

$$= \frac{24n}{n} - (3)^2 = 24 - 9 = 15$$

24. (b)

$$23, 29, 32, 49, 118$$

$$n = 5$$

$$P_i = \frac{i(n+1)}{100} = i \text{ th percentile}$$

$$P_{10} = \frac{10(n+1)}{100} = \frac{n+1}{10} = \frac{6}{10} = \frac{3}{5} \approx 1$$

$$P_{50} = \frac{50(n+1)}{100} = \text{median} = \frac{n+1}{2} = \frac{5+1}{2} = 3$$

$$\Rightarrow 23, 32$$

25. (c)

$$\text{Mean} = \frac{x_1 + x_2 + x_3 + \dots + x_{25}}{25} = 38$$

$$x_1 + x_2 + x_3 + \dots + x_{25} = 38 \times 25$$

23 is misread as 25 and 38 misread 36.

There is difference of $25 - 23 = 2$

and difference of $36 - 38 = -2$ in the sum.

Therefore there is no any difference in the net sum during calculation of mean.

Hence, the mean remain constant.

26. (c) Let the two numbers be x and y .

$$\text{Given, variance } \sigma^2 = 124$$

$$\text{Mean, } \bar{x} = 5 \text{ and } n = 5$$

$$\frac{1+2+6+x+y}{5} = 5$$





$$x + y = 16$$

So, mean deviation

$$= \frac{|1-5| + |2-5| + |6-5| + |x-5| + |y-5|}{5}$$

Now, we consider $x, y > 5$

$$\text{Mean deviation} = \frac{4+3+1+(x+y-10)}{5}$$

$$= \frac{8+16-10}{5} = \frac{14}{5} = 2.8$$

27. (c)

$$\bar{x} = 40 = \frac{\sum x_i}{100}$$

$$\Rightarrow \sum x_i = 4000$$

$$\sum x_i = 4000 - (52 + 53) + (25 + 35) = 3955$$

$$\Rightarrow \text{Correct } \bar{x} = 39.55$$

As from given options only (c) option is matched.

28. (d)

Class	Frequency	Cumulative Frequency
10 – 20	180	180
20 – 30	f_1	$180 + f_1$
30 – 40	34	$214 + f_1$
40 – 50	180	$394 + f_1$
50 – 60	136	$530 + f_1$
60 – 70	f_2	$530 + f_1 + f_2$
70 – 80	50	$580 + f_1 + f_2$

Total frequency = 685

$$580 + f_1 + f_2 = 685$$

$$f_1 + f_2 = 105$$

Median = 42.6 (given) lies in (40 – 50) interval

Class = 40 – 50

$$\text{Median} = l + \frac{\frac{N}{2} - C.f}{f} \times h$$

$$l = 40, h = 10$$

$$C.f = 214 + f_1$$

$$42.6 = 40 + \frac{\frac{685}{2} - (214 + f_1)}{180} \times 10$$

$$26 \times 18 = 3425 - 214 - f_1$$

$$f_1 = 817 \approx 82$$

$$f_2 = 105 - 82 = 23$$

29. (a) Mean, Median Mode = 1, 3^x , 9^x

Mode = 3 median – 2 mean

$$9^x = 3 \times 3^x - 2 \times 1$$

$$(3^x)^2 = 3 \times 3^x - 2$$

$$\text{Let } 3^n = x$$

$$x^2 - 3x + 2 = 0$$

$$(x + 1)(x - 2) = 0$$

$$x = 1, \quad 9^x = 1$$

$$x = 2, \quad 9^x = 4$$

Mode = 1, 4

30. (d) $m_1 = 2 \rightarrow A$

$$m_2 = 4 \rightarrow B$$

Median of combined series

$$M \geq \min (M_1, M_2)$$

$$M \geq 2$$

31. (d)

Total students = 40 (female) + 20 (male) = 60

Use the weighted average formula:

$$\text{Average height} = \frac{(40 \times 5.15) + (20 \times 5.66)}{60}$$

$$= \frac{206 + 113.2}{60}$$

$$= \frac{319.2}{60} = 5.32$$

32. (b)

(b) The variance is expressed in the same units as the units of observation

Reason:

Variance is in squared units of the original data (e.g., if data is in cm, variance is in cm^2).

Standard deviation (SD) is in the same units as the data, not variance.

